1	**(a)**	$g = GM/R^2$	C1	
		$= (6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(3.4 \times 10^6)^2 = 3.7\,\text{N kg}^{-1}$	A1	[2]
	(b)	$\Delta E_P = mg\Delta h$		
		because $\Delta h \ll R$ (or $1800\,\text{m} \ll 3.4 \times 10^6\,\text{m}$) g is constant	B1	
		$\Delta E_P = 2.4 \times 3.7 \times 1800$	C1	
		$= 1.6 \times 10^4\,\text{J}$	A1	[3]
		(*use of $g = 9.8\,\text{m s}^{-2}$ max. 1 for explanation*)		
	(c)	gravitational potential <u>energy</u> = $(-)GMm/x$	C1	
		$v^2 = 2GM/x$	C1	
		$x = 4D = 4 \times 6.8 \times 10^6$	C1	
		$v^2 = (2 \times 6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(4 \times 6.8 \times 10^6)$		
		$= 3.14 \times 10^6$		
		$v = 1.8 \times 10^3\,\text{m s}^{-1}$	A1	[4]
		(*use of 3.5D giving $1.9 \times 10^3\,\text{m s}^{-1}$, allow max. 3*)		